ECON 131 — Midterm 1

Key Concepts & Abbreviations Reference

Spring 2026 • Prof. Yagan • All 7 Lectures

Abbreviations & Acronyms

Abbrev.	Full Name	One-Sentence Summary
CBO	*Congressional Budget Office*	Nonpartisan federal agency that produces economic and budget projections for Congress.
CPI	*Consumer Price Index*	Measures inflation by tracking price changes in a fixed basket of goods; used to update the official poverty line but tends to overstate inflation.
CPI-RS	*CPI Research Series*	An improved version of the CPI that corrects for substitution bias; when used, it shows a larger decline in poverty than the official measure.
DD	*Difference-in-Differences*	An empirical estimator that identifies a causal effect by comparing the change over time in a treatment group to the change in a control group.
DWL	*Deadweight Loss*	The efficiency cost of a tax — transactions that would have been mutually beneficial but no longer happen because the tax makes them unprofitable.
EITC	*Earned Income Tax Credit*	A refundable U.S. tax credit for low-to-moderate income workers that rises then falls with earnings; the most important U.S. anti-poverty program for working-age adults.
FOC	*First-Order Condition*	The derivative of an objective function set equal to zero; the mathematical condition that characterizes the optimal choice in a maximization problem.
GDP	*Gross Domestic Product*	Total value of all goods and services produced in a country in a year; the standard measure of economic size.
MRS	*Marginal Rate of Substitution*	The rate at which a consumer is willing to trade one good for another while remaining equally happy; equals MU_1/MU_2 and equals the price ratio at the optimum.
MU	*Marginal Utility*	The additional utility gained from consuming one more unit of a good; typically diminishes as consumption increases.
PD	*Primary Deficit*	Government spending on programs minus tax revenue, excluding interest payments on existing debt; $PD = G - T$.
RCT	*Randomized Controlled Trial*	The gold-standard empirical method where units are randomly assigned to treatment and control, so any outcome difference is causally attributable to the treatment.
SNAP	*Supplemental Nutrition Assistance Program*	Federally funded food stamps; a non-cash transfer excluded from the official poverty rate measure.
TIE	*Taxable Income Elasticity*	The percentage change in taxable income in response to a 1% change in the after-tax wage; the key behavioral parameter for optimal tax analysis.

Core Concepts — All Lectures

Organized in lecture order (L01 → L07). Each entry is a 1–2 sentence plain-English summary.

L01 — Introduction to Public Economics

Concept	Plain-English Summary
Public Economics	The study of the government's role in the economy — when and how it should intervene, what the effects are, and why it does what it does.
Market Failure	A situation where the free market, left alone, produces an inefficient outcome; the primary economic justification for government intervention.
Externality	A cost or benefit imposed on third parties not involved in a transaction (e.g., carbon emissions); causes the market to over- or under-produce relative to the social optimum.
Adverse Selection	An asymmetric information problem where one party (e.g., the sick in an insurance market) self-selects in ways that unravel the market; a key market failure in health insurance.
Equity-Efficiency Trade-off	The tension between redistribution (equity) and productive incentives (efficiency) — taxing high earners to give to low earners may reduce total output.
Normative Economics	Analysis of how things *should* be; involves value judgments and policy recommendations. Requires positive analysis as a prerequisite.
Positive Economics	Analysis of how things actually *are*; empirical and testable, e.g., 'does a tax cut raise labor supply?' No value judgments involved.

L02 — Theoretical Tools

Concept	Plain-English Summary
Utility Function	A mathematical function that assigns a numerical score (utility) to every consumption bundle; captures preferences in a way that allows optimization.
Ordinal Utility	Utility that carries only preference ordering information, not magnitude — a bundle with utility 10 is preferred to one with utility 5, but not necessarily 'twice as good.'
Indifference Curve	The set of all consumption bundles that give a consumer the same utility level; slopes downward and higher curves represent more preferred bundles.
Diminishing Marginal Utility	The principle that each additional unit of a good provides less extra utility than the previous unit; explains why indifference curves are bowed inward.
Budget Constraint	The set of all consumption bundles a household can afford given its income and market prices; its slope equals the negative price ratio ($-p_1/p_2$).
Utility Maximization	Choosing the consumption bundle that reaches the highest indifference curve subject to the budget constraint; the solution condition is MRS = price ratio.
Substitution Effect	The change in behavior due purely to a relative price change, holding utility constant — e.g., a higher wage makes leisure relatively more expensive, pushing toward more work.
Income Effect	The change in behavior due purely to a change in purchasing power, holding relative prices constant — e.g., higher income leads to 'buying' more leisure, reducing work.

Concept	Plain-English Summary
Elasticity	A unit-free measure of responsiveness: the percentage change in one variable divided by the percentage change in another; used to compare sensitivity across markets.

L03 — Empirical Tools

Concept	Plain-English Summary
Counterfactual	The hypothetical outcome that would have occurred in the absence of a policy; causal inference is fundamentally about constructing a credible counterfactual.
Natural Experiment	A real-world situation where an external event creates arguably random variation in treatment, allowing causal identification without a formal RCT.
Parallel Pre-Trends	The key assumption of difference-in-differences: that without the policy, treatment and control groups would have trended the same way; validated by checking pre-period trends.
Causal Inference	Distinguishing correlation from causation in observational data; the central challenge in empirical public economics.
Omitted Variable Bias	Bias in an estimated causal effect due to unobserved factors that affect both treatment and outcome; the main threat that DD, RCTs, and natural experiments are designed to solve.

L04 — Budget & Debt Analysis

Concept	Plain-English Summary
Debt Dynamics Equation	$B(t+1) = (1+r)B_t + PD_t$; describes how government debt evolves as old debt accrues interest and new borrowing (the primary deficit) adds to the stock.
Primary Deficit	$G - T$; government spending minus tax revenue, excluding interest payments; reflects current fiscal policy choices, separate from the legacy cost of past debt.
Laffer Curve	The relationship between the tax rate and tax revenue; revenue is zero at both 0% and 100% and peaks at $\tau^* = 1/(1+\varphi)$ — beyond that, higher rates lose revenue.
Revenue-Maximizing Tax Rate	$\tau^* = 1/(1+\varphi)$; the tax rate at the peak of the Laffer curve; higher φ (more elastic taxable income) means a lower revenue-maximizing rate.
Sustainable Debt	A debt level that can be held constant over time; requires either a primary surplus or that the economic growth rate exceeds the interest rate ($g > r$).
r vs. g Condition	The condition $r < g$ (interest rate below growth rate) under which a government can sustain positive debt even while running primary deficits, because GDP grows faster than debt.

L05 — Income Distribution & Poverty

Concept	Plain-English Summary
Gini Coefficient	A summary measure of inequality ranging from 0 (perfect equality) to 1 (one person owns everything); the U.S. pre-tax Gini is ~0.50, post-tax ~0.38.
Lorenz Curve	A graph of the cumulative income share (y-axis) against the cumulative population share (x-axis); the Gini is the area between this curve and the 45-degree equality line.

Concept	Plain-English Summary
Official Poverty Rate	The share of the population with pre-tax cash income below the poverty threshold; criticized for using the flawed CPI and excluding non-cash transfers like the EITC.
EITC Phase-in Range	The earnings range where the EITC credit rises with earnings, effectively subsidizing work; creates a strong positive substitution effect that encourages labor force participation.
EITC Phase-out Range	The earnings range where the EITC credit falls as earnings rise, creating an implicit marginal tax rate on top of the statutory rate; can discourage additional work at the margin.
Capital Income	Income earned from owning assets (rk); far more concentrated than labor income — the top 1% own ~40% of U.S. private wealth — so rising capital shares worsen inequality.

L06 — Tax Incidence

Concept	Plain-English Summary
Statutory Incidence	Who the law says must pay a tax (who writes the check to the government); rarely the same as who actually bears the economic burden.
Economic Incidence	Who actually bears the burden of a tax in terms of reduced purchasing power; determined entirely by the relative elasticities of supply and demand, not by the legal assignment.
Tax Incidence Rule	The less elastic side of the market bears more of the tax burden; perfectly inelastic demand → consumers bear 100%; perfectly elastic supply → consumers bear 100%.
Deadweight Loss	The efficiency cost of a tax arising from transactions that no longer occur; rises with the square of the tax rate and with the elasticities of supply and demand.
Ramsey Rule	The principle that for efficient taxation, goods with more inelastic demand should be taxed at higher rates, because inelastic goods have lower deadweight loss per dollar of revenue.
Danish High-Earner Study	Kleven, Landais & Saez (2014): a 1991 Danish tax break for high-earning foreigners caused a sharp increase in their migration to Denmark, showing that top earners are highly mobile in response to taxes.

L07 — Optimal Labor Tax & Labor Supply

Concept	Plain-English Summary
Labor Supply Function	$l^* = [(1-t)w]^\eta$; the optimal hours of work chosen by a household given the after-tax wage and the disutility parameter η.
Taxable Income Elasticity	In the standard model, TIE = η exactly; measures how much labor supply (earnings) changes in percentage terms for a 1% change in the after-tax wage.
Lump-Sum Transfer	A fixed government payment that does not depend on any household choice; has only an income effect (no substitution effect) and creates no deadweight loss.
Optimal Top Tax Rate	The tax rate that maximizes social welfare trading off redistribution gains against efficiency costs; lower when taxable income elasticity is higher because behavioral responses are larger.

Concept	Plain-English Summary
Utilitarian Social Welfare	A social objective that maximizes the sum of all individuals' utilities; with diminishing marginal utility and no behavioral response to taxes, this implies full consumption equalization.
Labor Market Equilibrium	The wage w* at which labor supply equals labor demand; a tax reduces w* — the burden is shared between workers and firms depending on both supply (η) and demand (η^d) elasticities.

Key Formulas at a Glance

Concept	Formula	Note
Debt dynamics	$B(t+1) = (1+r)B_t + PD_t$	*$PD_t = G_t - T_t$ (primary deficit)*
Tax revenue (Laffer)	$TR(\tau) = \tau(1-\tau)^\varphi$	*Taxable income = $(1-\tau)^\varphi$*
Revenue-max tax rate	$\tau^* = 1 / (1 + \varphi)$	*Peak of Laffer curve*
Sustainable debt	$B = (TR - G) / r$	*Constant B, G, τ; requires TR > G or g > r*
Budget constraint (HH)	$c = (1-t)wl + T$	*After-tax earnings plus lump-sum transfer*
Optimal labor supply	$l^* = [(1-t)w]^\eta$	*FOC of standard utility model*
Optimal consumption	$c^* = [(1-t)w]^{(1+\eta)} + T$	*Plug l* into budget constraint*
Taxable income elasticity	$TIE = \eta$	*% change in l* per 1% change in (1−t)w*
Equilibrium wage	$w^* = (1-t)^{(-\eta / (\eta + \eta^d))}$	*Set labor supply = labor demand, solve for w*
Consumer incidence share	$\varepsilon_s / (\varepsilon_s - \varepsilon_D)$	*$\varepsilon_s > 0$, $\varepsilon_D < 0$; less elastic side bears more*
DWL (conceptual)	$\propto (\Delta t)^2 \times \text{elasticity}$	*Rises with square of tax rate and elasticity*